HPGAN: Hyperspectral Pansharpening Using 3-D Generative Adversarial Networks

Weiying Xie, *Member, IEEE*, Yuhang Cui, Yunsong Li, Jie Lei, *Member, IEEE*, Qian Du, *Fellow, IEEE*, and Jiaojiao Li, *Member, IEEE*

***Abstract*— Hyperspectral (HS) pansharpening, as a special case of the superresolution (SR) problem, is to obtain a high-resolution (HR) image from the fusion of an HR panchromatic (PAN) image and a low-resolution (LR) HS image. Though HS pansharpening based on deep learning has gained rapid development in recent years, it is still a challenging task because of the following requirements: 1) a unique model with the goal of fusing two images with different dimensions should enhance spatial resolution while preserving spectral information; 2) all the parameters should be adaptively trained without manual adjustment; and 3) a model with good generalization should overcome the sensitivity to different sensor data in reasonable computational complexity. To meet such requirements, we propose a unique HS pansharpening framework based on a 3-D generative adversarial network (HPGAN) in this article. The HPGAN induces the 3-D spectral–spatial generator network to reconstruct the HR HS image from the newly constructed 3-D PAN cube and the LR HS image. It searches for an optimal HR HS image by successive adversarial learning to fool the introduced PAN discriminator network. The loss function is specifically designed to comprehensively consider global constraint, spectral constraint, and spatial constraint. Besides, the proposed 3-D training in the high-frequency domain reduces the sensitivity to different sensor data and extends the generalization of HPGAN. Experimental results on data sets captured by different sensors illustrate that the proposed method can successfully enhance spatial resolution and preserve spectral information.**

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Weiying Xie, Yuhang Cui, Yunsong Li, Jie Lei, and Jiaojiao Li are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: wyxie@xidian.edu.cn; ysli@mail.xidian.edu.cn; jielei@mail.xidian.edu.cn; jjli@xidian.edu.cn).

Qian Du is with the Department of Electronic and Computer Engineering, Mississippi State University, Starkville, MS 39759 USA (e-mail: du@ece.msstate.edu).

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***Index Terms*— Generative adversarial networks, global constraint, hyperspectral pansharpening, spectral–spatial constraints, 3-D high-frequency block.**

I. Introduction

HYPERSPECTRAL (HS) image is a data cube that consists of a series of 2-D spectral channels, and each channel characterizes the reflectance or radiance of a scene in a certain narrow wavelength range. The vast amount of spectral information regarding the physical nature of material substances has achieved significant success for land-use classification [1]–[3], target detection [4], [5], and denoising [6]. As a consequence, high-resolution (HR) HS images are required to facilitate fast and accurate remote sensing interpretation. However, it remains a challenging problem to capture HS images at the desired resolution because of economic, technical, or physical limitations. The superresolution (SR) technique is an economical way to create an HR HS image from a given low-resolution (LR) HS image, which is in a software manner to achieve the desired resolution without modifying the hardware. So far, the methods for superresolving HS images can be broadly categorized into single-image-based SR method and fusion-based SR method according to the input source. HS pansharpening is essentially one of the fusion-based SR problems where a related HR panchromatic (PAN) image of the same scene is used [7]. Compared with the single-image-based SR method, the HS pansharpening method achieves more accurate SR results with the auxiliary HR PAN image.

So far, the developed pansharpening methods can be roughly classified as component substitution (CS), multiresolution analysis (MRA), the Bayesian methods, matrix factorization methods, tensor-based methods, and deep learning-based methods [8], [9]. The CS-based methods utilize a PAN image to replace the spatial component of HS image by the inverse transform, which may bring about spectral distortion. The Gram–Schmidt (GS) and adaptive GS (GSA) [10] are representative of the CS-based methods. The MRA-based methods aim at injecting the spatial details of a PAN image into each band of the upsampled HS image. The representative MRA-based methods contain modulation transfer function (MTF), MTF generalized Laplacian pyramid (MTF-GLP) [11], MTF-GLP with high pass modulation (MTF-GLP-HPM) [12], and smoothing filter-based intensity modulation (SFIM) [13]. Spatial distortion such as ringing artifacts may be introduced

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by the MRA-based methods, but spectral preservation is better achieved in this type of method. The combination of the CS and MRA-based methods is a tradeoff between spatial SR and spectral preservation in which the guided filter PCA (GFPCA) [14] method won the "Best Paper Challenge" in the 2014 IEEE Data Fusion Contest. The Bayesian methods depend on a posterior distribution of the reference HR HS image to fuse LR HS and HR PAN images. The Bayesian sparsity promoted the Gaussian prior (Bayesian Sparse) [15], HySure [16], the Bayesian naive Gaussian prior (Bayesian Naive) [17], and a fast fusion method based on Sylvester equation (FUSE) [18] belong to this category. The matrix factorization methods alternately factorize the HR multispectral (MS) image and the LR HS image to obtain high-quality endmember and abundance maps from HS and MS images. The coupled nonnegative matrix factorization (CNMF) [19] and Lanaras method are the representatives of the matrix factorization pansharpening algorithms [20]. The tensor-based methods regard an HS image as a 3-D tensor and perform well, but they have high computational complexity [21], [22]. Within the last group, pansharpening solutions based on deep learning are recently promoted by the success of deep learning models but are still in the initial stage of exploration [23]. Among them, deep HS image sharpening (DHSIS) [24] adopted the deep priors learned by a convolutional neural network (CNN) into the HS–MS fusion framework. Li *et al*. [25] employed an SR model based on CNN to upsample the LR HS image in a band-by-band manner and then injected details, which is named as detailed-based deep Laplacian pansharpening (DDLPS). After that, Xie *et al*. [26] proposed an HS pansharpening method with deep priors (HPDP), which introduced the deep priors learned by a high-frequency net to solve the HS-PAN fusion problem. It is worth noting that the HS pansharpening method, i.e., HS-PAN fusion, is more difficult than HS-MS fusion due to the limited spectral information in a single-band PAN image. Either the HS-MS framework or the HS-PAN framework based on CNNs realizes spatial SR of 2-D spectral channels without considering the effect of deep learning on spectral preservation.

To achieve spatial SR and spectral preservation in a unified framework essentially, we propose a novel 3-D framework for HS pansharpening in this article. In particular, generative adversarial network (GAN) [27] is a very promising approach for single-image SR, and we specially develop it to be more in line with HS pansharpening for the first time. Note that the LR HS and HR PAN images to be fused have different dimensions. Based on this observation, we first generate an effective 3-D block from the LR HS image, and the newly constructed 3-D PAN cube to learn a 3DGAN. Considering the issue of spatial SR as well as spectral preservation, we employ 3-D convolutional layers to achieve the joint spectral and spatial learning. Specifically, the proposed HS pansharpening using a 3DGAN (HPGAN) model contains 3-D spectral–spatial generator network and PAN discriminator network. It can obtain an optimal HR HS image by the proposed 3-D spectral–spatial generator network with adversarial learning from the proposed PAN discriminator. In the 3-D spectral–spatial generator, the proposed loss function consists of global constraint, spectral constraint, and spatial constraint, where the spatial constraint is trained by adversarial learning. Besides, the reconstruction model is learned in a high-frequency domain to ensure a good generalization of the model. In the PAN discriminator, the simulated PAN image can be fully transformed from the recovered HR HS image by the generator and is distinguished from the real PAN image. By the continuously adversarial learning, the 3-D spectral–spatial generator network recovers HR HS image to produce more artificial PAN images, while the PAN discriminator has a stronger ability to make a distinction between the artificial and real PAN images. To verify the effectiveness of the proposed method, we train the model on part of a ground-based data set and test on five other data sets that are captured by different sensors operated at different wavelengths.

The main contributions of this article are summarized as follows.

1) We propose a novel HPGAN method that focuses on the issue of spatial SR and spectral preservation in a unique 3DGAN model in which all the parameters are adaptively trained by adversarial learning without human intervention. To the best of our knowledge, there is no other work in regard to 3DGAN in HS pansharpening.
2) A 3-D spectral–spatial generator network is induced to reconstruct the HR HS image from the newly constructed 3-D PAN cube and the LR HS image, which realizes the nonlinear mapping relationship between two images of different dimensions.
3) Different from the existing methods that distinguish superresolved images from real images, PAN discriminator is introduced based on the spectral response function of HS imaging and learned to discriminate the simulated PAN images from the real one.

The remainder of this article is organized as follows. Section II reviews GANs. The proposed HPGAN model is introduced in Section III. Section IV presents experimental results. Section V draws the conclusions.

II. Related Work on GANs

Goodfellow *et al*. [27] presented the original GAN model that consists of a generator G and a discriminator D. With the goal of one beating the other, G aims at generating data that are close to the real data, while D is designed to distinguish real from generated data. By the successive adversarial learning, G generates more and more real data, while D has a stronger ability to distinguish. The objective of the original GAN model can be expressed as

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log(D(x))\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log(1-D(G(z)))\right] \quad (1)$$

where x denotes the real data drawn from the real data distribution $x \sim p_{\text{data}}(x)$, z represents a random input drawn from a probability distribution, such as the Gaussian $z \sim p_z(z)$, and $\mathbb{E}$ is the empirical estimate of the probability expectation. Because of the advantages of GANs, researchers have gained encouraging performance in many applications, especially in the field of image SR.

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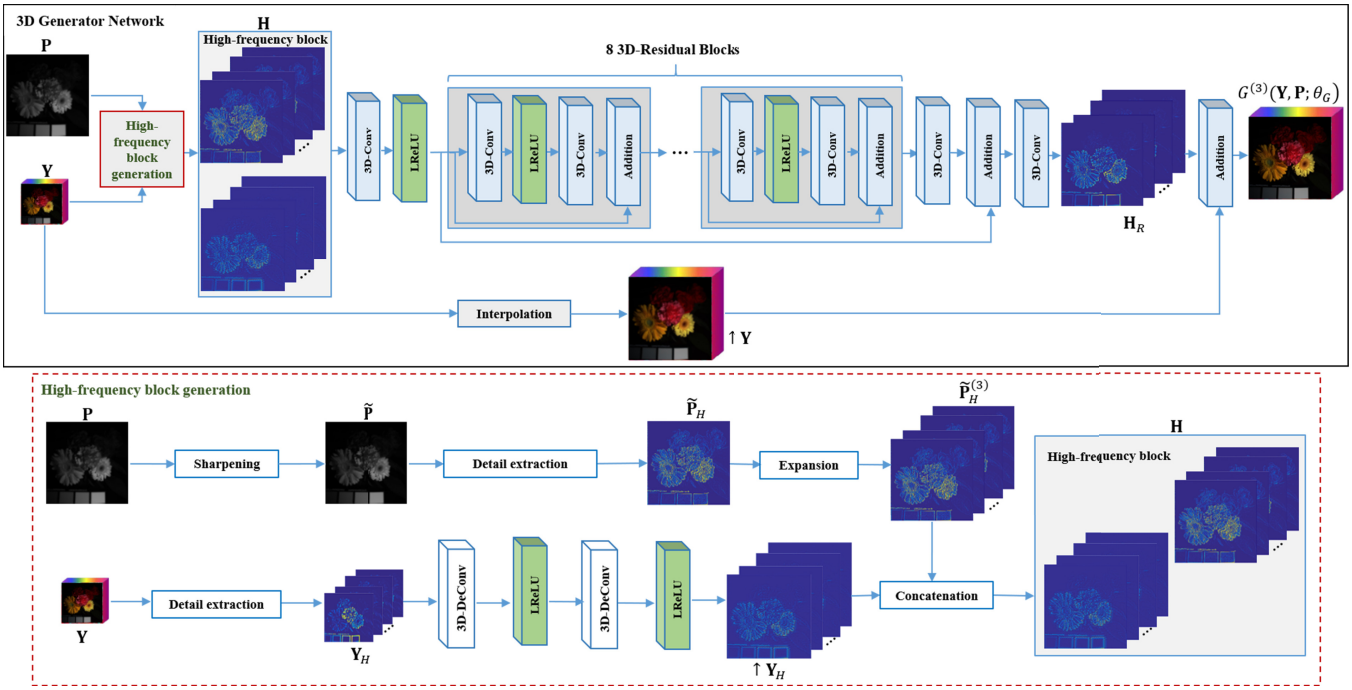


Fig. 1. Overall flowchart of the proposed 3-D spectral–spatial generator network.

Ledig *et al.* [28] explored GANs for image SR, named SRGAN, in which the generator G aims at generating an HR RGB image I_{SR} from a single LR image I_{LR}, while the discriminator D needs to distinguish the generated I_{SR} from the real HR image I_{HR}. The adversarial min–max problem for image SR can be represented as

$$\min_{\theta_G}\max_{\theta_D}\mathbb{E}_{I^{\mathrm{HR}}\sim p_{\mathrm{data}}(I^{\mathrm{HR}})}\left[\log\left(D_{\theta_D}\left(I^{\mathrm{HR}}\right)\right)\right] + \mathbb{E}_{I^{\mathrm{LR}}\sim p_G(I^{\mathrm{LR}})}\left[\log\left(1-D_{\theta_D}\left(G_{\theta_G}\left(I^{\mathrm{LR}}\right)\right)\right)\right]. \quad (2)$$

Just like SRGAN learns a generator to produce an HR image that fools a discriminator, the enhanced SRGAN (ESRGAN) adopted the residual-in-residual dense block (RRDB) without batch normalization (BN) [29] as the basic architecture [30]. It was indicated in ESRGAN that removing BN layers in a deeper network could increase SR performance and reduce computational complexity. Thus, we also remove BN layers in our HPGAN model. Besides, ESRGAN used the features before activation to improve the perceptual loss for better brightness consistency and texture recovery.

III. Proposed HPGAN Model

Let $\mathbf{X}\in\mathbb{R}^{W\times H\times L}$ be the desired HR HS image, where $W\times H$ is the number of spatial pixels and L is the number of spectral channels. The objective is to reconstruct $\mathbf{X}$ by the combination of an LR HS image $\mathbf{Y}\in\mathbb{R}^{w\times h\times L}$ and an HR PAN image $\mathbf{P}\in\mathbb{R}^{W\times H\times 1}$ of the same scene, where $w\times h=(W\times H)/s^2$ is the number of spatial pixels in $\mathbf{Y}$ and s denotes the scale factor. $\mathbf{P}$ only contains one channel, but the number of spatial pixels in $\mathbf{X}$ and $\mathbf{P}$ is the same. Instead of separately carrying out SR and spectral preservation as in [25] and [26], we build a comprehensive and adaptive pansharpening framework to superresolve HS image while preserving spectral information. It is worth noting that the 3-D spectral–spatial generator network is proposed to consider both spectral preservation and spatial SR, and a PAN discriminator is further defined to optimize the adversarial min–max problem

$$\min_{\theta_G}\max_{\theta_D}\mathbb{E}_{\mathbf{P}\sim q_{\mathrm{real}}(\mathbf{P})}[D(\mathbf{P};\theta_D)]^2 + \mathbb{E}_{\mathbf{Z}\sim q_G(\mathbf{Y},\mathbf{P})}\left[1-D\left(\mathbf{R}\cdot G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G);\theta_D\right)\right]^2. \quad (3)$$

Inspired by Mao *et al.* [31], the sigmoid cross-entropy loss function used in SRGAN and ESRGAN is replaced with the least-squares loss function. The proposed HPGAN model enables to learn the 3-D spectral–spatial generator network $G^{(3)}$ with the objective of deceiving the proposed PAN discriminator D that is learned to distinguish the synthetic PAN image from the real PAN image where the synthetic PAN image is the spectral response function (SRF) $\mathbf{R}$ multiplied by the estimated HR HS image, i.e., $\mathbf{Z}=\mathbf{R}\cdot G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)$. Here, $G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)$ represents the reconstructed HR HS image obtained by the proposed 3-D spectral–spatial generator network $G^{(3)}$ with the trained parameters θ_G. Concretely, the synthetic PAN image completely close to the real PAN image means that the HR HS image estimated by $G^{(3)}$ is similar to the reference image, and thus, it is difficult to be discriminated against by the proposed PAN discriminator D.

A. 3-D Spectral–Spatial Generator Network

The architecture of the proposed 3-D spectral–spatial generator network is illustrated in Fig. 1. With the input as LR HS image $\mathbf{Y}$ and HR PAN image $\mathbf{P}$, the proposed network aims at recovering the corresponding HR HS image $G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)$, where θ_G represents the whole parameters learned by $G^{(3)}$. Following our previous work [26] and PanNet [32], high-frequency learning has better generalization

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than image domain learning. Therefore, a 3-D high-frequency block, i.e., $\mathbf{H} = H_{\theta_H}(\mathbf{Y}, \mathbf{P}; \theta_H)$, is created in which the parameter θ_H is learned by the deconvolution layers. The detailed generation process of the 3-D high-frequency block is shown in the lower position of Fig. 1. Obviously, the high-frequency of the LR HS image is upsampled by the deconvolution layers rather than bicubic interpolation because the deconvolution layers can not only upsample high-frequency but also extract features from high frequency, which allows high-frequency learning from the very beginning. Besides, the parameters of the deconvolution layers θ_H are also updated by the loss function of our HPGAN model. To achieve 3-D training, the high-frequency of PAN image is expanded and concatenated with the upsampled high-frequency of LR HS image, resulting in the high-frequency block. Hence, the concatenation $[\tilde{\mathbf{P}}_H^{(3)}, \uparrow \mathbf{Y}_H]_{\theta_H}$ of $\tilde{\mathbf{P}}_H^{(3)}$ and $\uparrow \mathbf{Y}_H$ constitutes the sufficient samples to learn the proposed 3-D spectral–spatial generator network $G^{(3)}$, where $\tilde{\mathbf{P}}_H^{(3)}$ represents the 3-D high-frequency of PAN image and $\uparrow \mathbf{Y}_H$ is the upsampled high frequency of LR HS image. To estimate the pansharpened HR HS image $G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G)$, it is trained as

$$G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G) = f_{\theta_R}\big(H_{\theta_H}(\mathbf{Y}, \mathbf{P}; \theta_H); \theta_R\big) + \uparrow \mathbf{Y} \tag{4}$$

where $\uparrow \mathbf{Y}$ is the upsampled image of LR HS image obtained by bilinear interpolation, and $\mathbf{H}_R = f_{\theta_R}(\mathbf{H}; \theta_R)$ is the learned deep details from the high-frequency block, as demonstrated in Fig. 1. In summary, the whole parameter θ_G is the combination of θ_H and θ_R, i.e., $\theta_G = \{\theta_H, \theta_R\}$. It is worth noting that all the parameters are adaptively learned by the 3-D spectral–spatial generator $G^{(3)}$ without human intervention, which is one of the key contributions in our HPGAN model. In particular, there are four types of layers in the proposed 3-D spectral–spatial generator network $G^{(3)}$ with 21 layers, as illustrated in Fig. 1.

1) $3D - DeConv + LReLU$: In the high-frequency block generation process, the resolution of the input high-frequency $\mathbf{Y}_H$ is enhanced by two trained 3-D deconvolution layers of size $6 \times 6 \times 6 \times 1$, and then, the leaky rectified linear units (LReLU, $\max(0.2x, x)$) are used.
2) $3D - Conv + LReLU$: Thirty two filters of size $3 \times 3 \times 3 \times 2$ are used to generate 32 3-D feature maps, and LReLUs are also employed for nonlinearity. Here, 2 represents the input high-frequency block that is a combination of PAN and HS images.
3) $3D - Conv + LReLU + 3D - Conv$: For layers 4–19, 32 filters of size $3 \times 3 \times 3 \times 32$ are used, and LReLU is added between two adjacent convolution layers. With skip connection, it constitutes the 3-D residual block that has a residual-in-residual structure. Here, residual learning is used on different levels.
4) $3D - Conv$: For layer 20, 32 filters of size $3 \times 3 \times 3 \times 32$ are used to remap the residuals learned by eight 3-D residual blocks; for the last layer, a filter of size $3 \times 3 \times 3 \times 1$ is used to reconstruct the output.

B. PAN Discriminator Network

As mentioned earlier, by the successive adversarial learning, the generator generates more and more real data, while the discriminator has a stronger ability to distinguish. The proposed discriminator D is learned to distinguish the synthetic PAN image from real PAN image, which is significantly different from SRGAN and ESRGAN that aim at distinguishing superresolved images from real images. It is worth noting that if $G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G) \approx \mathbf{X}$, then $\mathbf{R} \cdot G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G) \approx \mathbf{R} \cdot \mathbf{X}$. If the discriminator cannot distinguish the synthetic PAN image generated by generator result from real PAN image, the HS pansharpening result is better. Based on this observation, we propose the PAN discriminator by a 2-D CNN, which has fewer parameters, smaller models, and lower feature dimensions than a 3-D discriminator network. Meanwhile, its performance is stable and acceptable in the course of the experiment. Specifically, the discriminator loss l_D is redefined as

$$l_D = (D(\mathbf{P}; \theta_D) - 1)^2 + D\big(\mathbf{R} \cdot G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G); \theta_D\big)^2. \tag{5}$$

The PAN discriminator outputs a scalar between 0 and 1 indicating the probability. When $D(\mathbf{P}; \theta_D)$ approximates 1 and $D(\mathbf{R} \cdot G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G); \theta_D)$ approximates 0, l_D reaches its minimum. The architecture of the proposed PAN discriminator network is designed, as shown in Fig. 2. D has eight 2-D convolution layers and two fully connected layers, and the filter size of each convolution layer is 3×3. Besides, LReLU is used as the activation function. In Fig. 2, "n32s1" means that 32 filters are used to extract 32 feature maps, and the stride is 1, similarly for "n32s2," "n64s1," "n64s2," "n128s1," "n128s2," "n256s1," and "n256s2." The resulting 256 feature maps are followed by two dense layers to obtain a probability for sample discrimination.

C. 3-D Spectral–Spatial Generator Loss

The 3-D spectral–spatial generator loss function $l_{G^{(3)}}$ is the key design to obtain an ideal HR HS image. Besides encouraging the loss to follow the minimization of the mean squared error (MSE) in SRGAN, the angular difference of the reference HR HS image $\mathbf{X}$ and the reconstructed HR HS image $G^{(3)}(\mathbf{Y}, \mathbf{P}; \theta_G)$ is further minimized. In this article, spectral angle mapper (SAM) [33] is added to constrain the 3-D spectral–spatial generator network. During the optimization procedure, an adversarial loss component is also proposed to fool the PAN discriminator network. Thus, the proposed 3-D spectral–spatial generator can be formulated to minimize the overall loss defined as

$$l_{G^{(3)}} = l_{G^{(3)}}^{\text{MSE}} + \alpha_1 l_{G^{(3)}}^{\text{SAM}} + \alpha_2 l_{G^{(3)}}^{\text{ADV}}. \tag{6}$$

It is worth noting that the generator loss fully considers global constraint, spectral constraint, and spatial constraint. $l_{G^{(3)}}^{\text{MSE}}$ is a global constraint to make all the pixels that are generated by $G^{(3)}$ approximate to all the pixels in the reference HS image, including $W \times H$ pixels in the spatial domain and L points in the spectral domain of each pixel. As mentioned earlier, $l_{G^{(3)}}^{\text{SAM}}$ represents the spectral constraint whose purpose is to minimize the spectral angular distance between the reference spectral vector and the reconstructed one. There are $W \times H$ spectral vectors for each HS image. In fact, $l_{G^{(3)}}^{\text{ADV}}$ is a spatial constraint, which aims at generating an approximate real PAN

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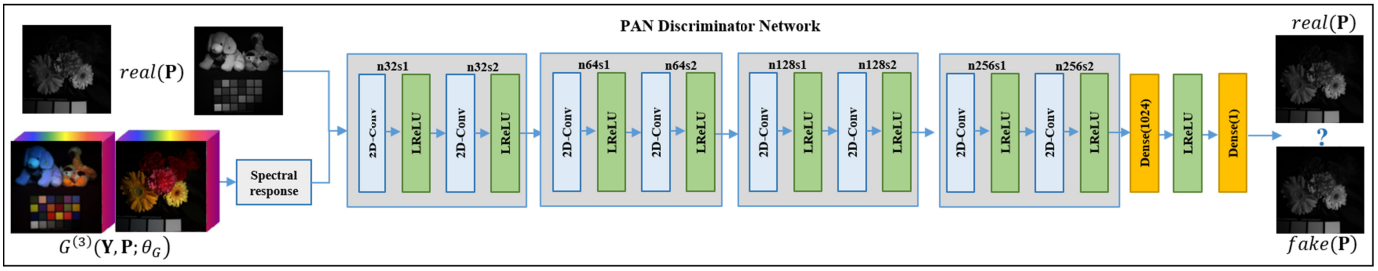


Fig. 2. Overall flowchart of the proposed PAN discriminator network.

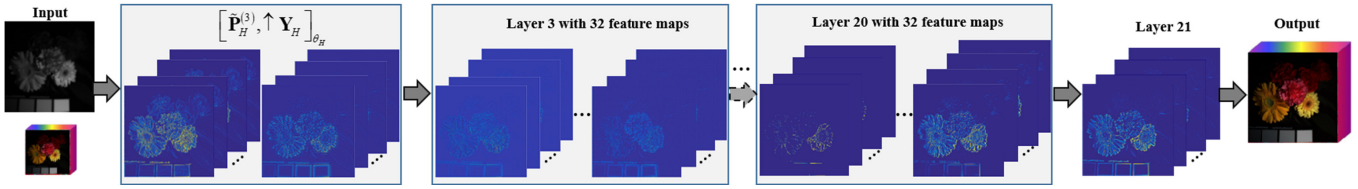


Fig. 3. Example feature maps of the proposed 3-D spectral–spatial generator network.

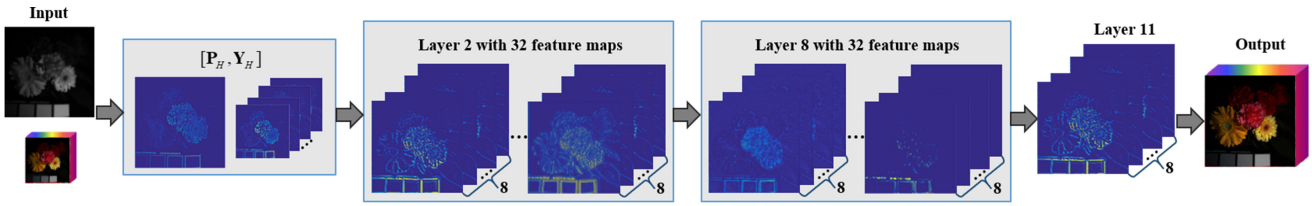


Fig. 4. Example feature maps of the 2-D network.

image to fool the proposed PAN discriminator. It is precisely to ensure spatial structure, we have innovatively proposed the PAN discriminator network. As a consequence of the abovementioned comprehensive loss function, our proposed 3-D spectral–spatial generator can generate HR HS images with spectral preservation. In our experiments, $\alpha_1 = 10^{-4}$ and $\alpha_2 = 10^{-3}$. Concretely, the 3-D MSE loss $l_{G^{(3)}}^{\text{MSE}}$ is defined as

$$l_G^{\text{MSE}} = \frac{1}{WHL}\sum_{i=1}^{W}\sum_{j=1}^{H}\sum_{k=1}^{L}\left(\mathbf{X}_{i,j,k} - G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)_{i,j,k}\right)^2 \quad (7)$$

and the adopted spectral loss based on SAM can be represented as

$$l_G^{\text{SAM}} = \frac{1}{WH}\sum_{i=1}^{W}\sum_{j=1}^{H}\arccos\frac{G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)_{i,j}\mathbf{X}_{i,j}^T}{\|G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G)_{i,j}\|_2 \,\|\,\mathbf{X}_{i,j}\|_2}. \quad (8)$$

The purpose of adversarial learning is to make the PAN discriminator unable to distinguish real PAN and synthetic PAN images. Therefore, the following adversarial loss function to make $D(\mathbf{R}\cdot G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G);\theta_D)$ approach its maximum value is defined as

$$l_G^{\text{ADV}} = \left(D\left(\mathbf{R}\cdot G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G);\theta_D\right) - 1\right)^2. \quad (9)$$

The maximization of l_G^{ADV} means that $D(\mathbf{R}\cdot G^{(3)}(\mathbf{Y},\mathbf{P};\theta_G);\theta_D)$ is approximately equal to $D(\mathbf{P};\theta_D)$, which is the probability of PAN discriminator with the real PAN image as the input.

D. Implementation Details

The filter size of each convolutional layer is set as $3 \times 3 \times 3$, and each deconvolution layer is defined as $6 \times 6 \times 6$ in the proposed HPGAN. LReLU with 0.2 slope on negative inputs is employed as the activation function. To ensure that the feature maps are of the same size as the input, all the convolution layers are given sufficient zero-padding around the boundaries.

It is worth noting that the model is trained through a subset of a public HS data set. There are 32 HS images in the CAVE data set[1] [34] in the wavelength range 400–700 nm with 10 nm a band. The HS images in this data set contain a wide variety of real-world materials and objects with controlled illumination in the laboratory environment. Each HS image contains 31 spectral bands with 512×512 pixels in the spatial domain. The first 20 (60%) HS images in CAVE are selected as training samples, and the remaining 12 HS images are used as testing samples. Limited by memory and insufficient samples in HS images, our training is patch-based. In particular, the patches of size $128 \times 128 \times 31$ are extracted from each training HS image ($512 \times 512 \times 31$) in CAVE resulting 16 patches as reference per HS image. In this way, a total number of 320 (20×16) patches are used for training, and each patch contains 31 spectral bands. Following DHSIS and HPDP, the LR HS image is constructed by applying a Gaussian filter and then downsampled with a scaling factor

[1]http://www.cs.columbia.edu/CAVE/databases/multispectral/

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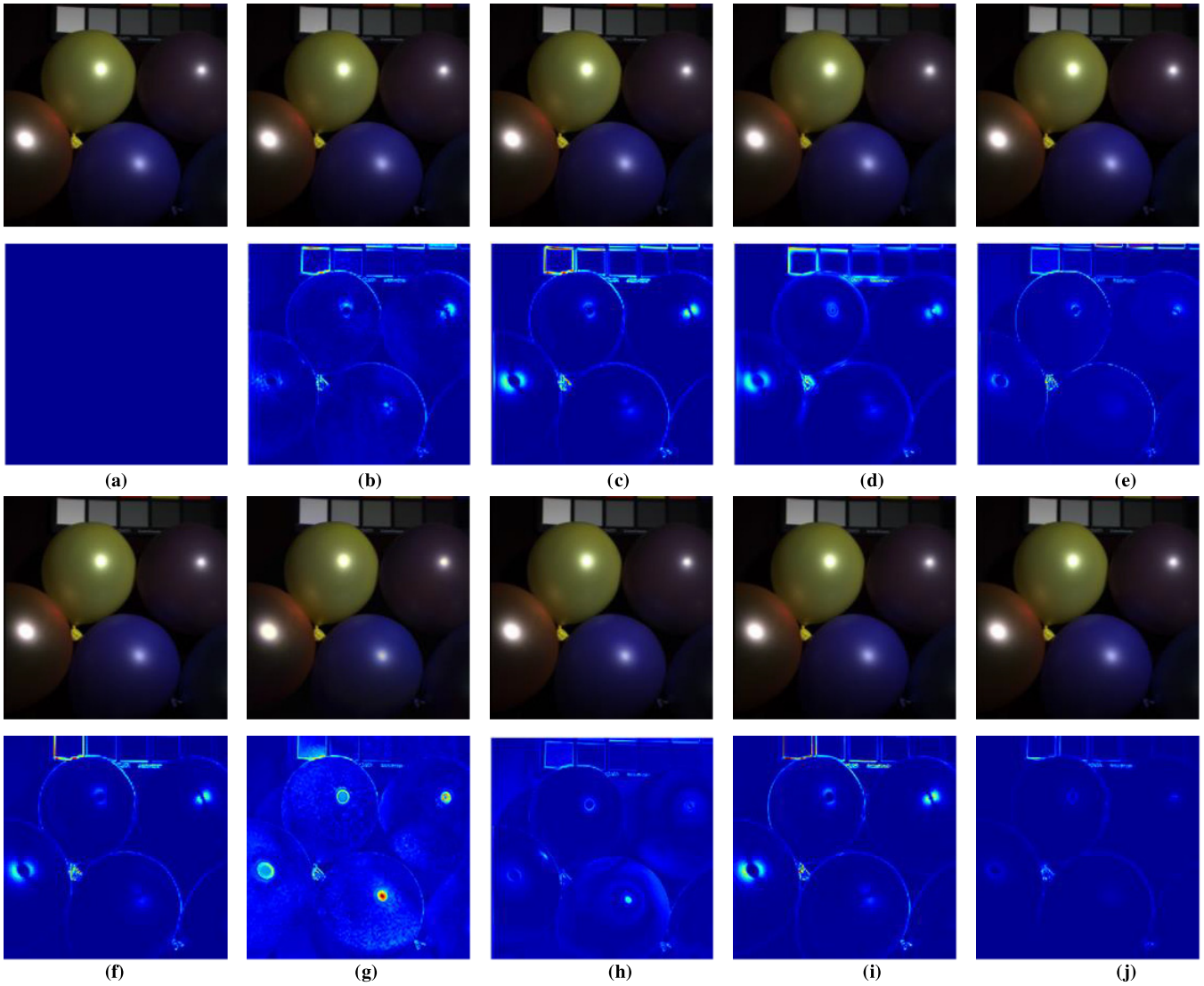


Fig. 5. Visual results obtained by different methods on one HS image from the CAVE data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false color image is chosen for clear visualization (red: 30, green: 20, and blue: 10). The even rows show the difference maps of the corresponding method.

of 4. The HR PAN image is stimulated by downsampling the reference with the spectral model. In the 3-D spectral–spatial generator training batch, four HS-PAN patches are randomly sampled with the sizes of $32 \times 32 \times 31$ for HS image and 128×128 for PAN image. In the PAN discriminator training batch, real and simulated PAN patches with the size of 128×128 are randomly selected. The Adam optimization is used for the generator and discriminator with 2000 training epochs, and each epoch has 80 iterations. Besides, the learning rate is initialized to 10^{-5} for all layers and decreased to 10^{-6} after 1000 epochs.

IV. Experimental Results and Analysis

A. Compared Methods and Quantitative Metrics

For comprehensive validation, the proposed HPGAN method is compared with several methods that are representative of the categories reviewed in Section I and also show competitive performance. These methods are GSA, MTF-GLP, GFPCA, FUSE, HySure, CNMF, Lanaras, and DDLPS.

In addition to subjective visual comparison, five evaluation indexes are used for objective assessment, including peak SNR (PSNR), cross correlation (CC), SAM [33], root-mean-squared error (RMSE), and Erreur Relative Globale Adimensionnelle de Synthése (ERGAS) [35]. The CC and SAM measure spatial and spectral qualities, respectively. The better pansharpening result means the larger CC and the smaller SAM. The RMSE and ERGAS measure the global quality of the pansharpened HS image, and the optimal value of RMSE and ERGAS is 0.

B. Mid-Level Representation

Fig. 3 shows some feature maps for one HS image from the CAVE data set extracted by the 3-D spectral–spatial generator network at different layers. The 3-D generator first produces a 3-D high-frequency block that is further convoluted by 3-D

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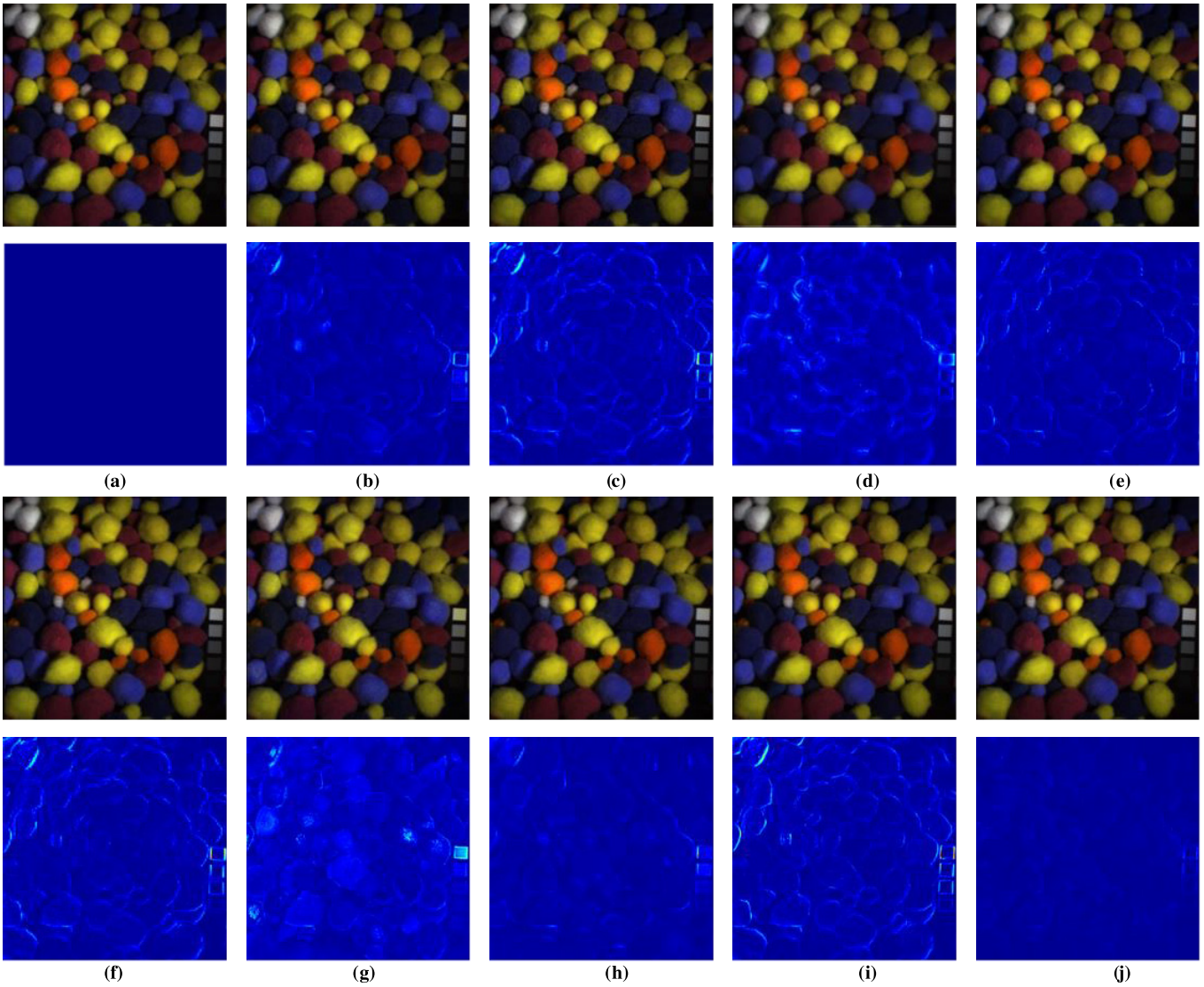


Fig. 6. Visual results obtained by different methods on another HS image from the CAVE data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false color image is chosen for clear visualization (red: 30, green: 20, and blue: 10). The even rows show the difference maps of the corresponding method.

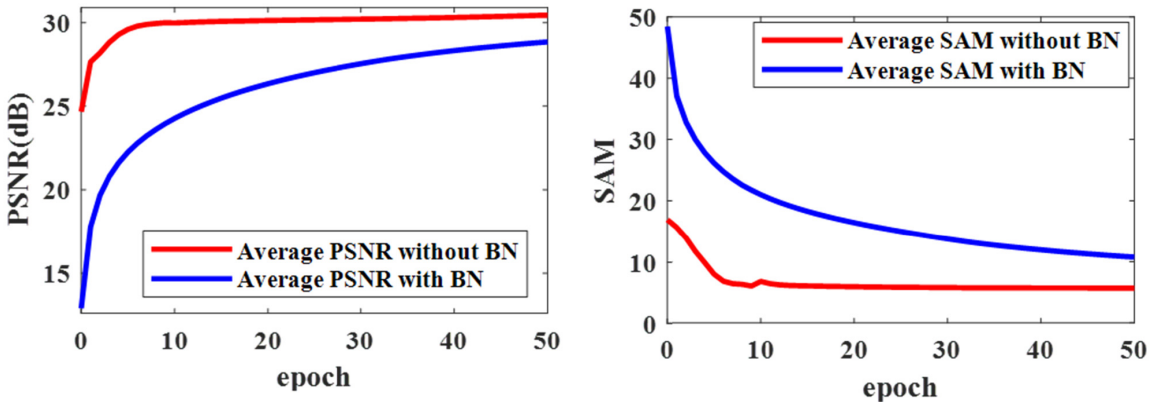


Fig. 7. Effects of BN layers on PSNR and SAM with respect to epochs.

kernels. It is clear that the proposed network can accurately extract 3-D features of the high-frequency block, which can be further used to recover the HR HS images. Fig. 4 shows some feature maps extracted by the 2-D network. The significant difference between the 3-D network and 2-D network is that the input of the 2-D network is each band (or some bands)

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TABLE I
OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE CAVE DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9920	5.1146	0.0145	4.6386	36.5333
MTF-GLP	0.9937	3.7704	0.0131	4.1423	37.3072
GFPCA	0.9936	4.1959	0.0133	4.2588	37.2495
HySure	0.9930	6.3357	0.0121	5.0278	38.0415
FUSE	0.9930	4.4552	0.0135	4.3927	37.1224
Lanaras	0.9844	8.5395	0.0191	5.8098	35.2669
CNMF	0.9935	5.2907	0.0118	4.1435	38.3981
DDLPS	0.9923	6.9385	0.0144	4.6592	36.4901
HPGAN	**0.9981**	**2.8019**	**0.0065**	**2.1608**	**43.6114**

of HS image rather than a complete HS of the 3-D network. It is worth noting that the 3-D network considers both spatial information and spectral information, while the 2-D network does not.

C. Experimental Results

Table I reports the average evaluation indexes of the competitive methods on 12 generated HR HS images (testing data) of the CAVE data set. For clarity, bold indicates the best results. It apparently indicates that the proposed HPGAN method achieves the best results and significantly outperforms the second one in terms of the average CC, SAM, RMSE, ERGAS, and PSNR. In particular, the average CC, SAM, RMSE, ERGAS, and PSNR obtained by the proposed HPGAN method are 0.9981, 2.8019, 0.0065, 2.1608, and 43.6114, which are much higher than those calculated by the second-best approach, 0.9937 (MTF-GLP), 3.7704 (MTF-GLP), 0.0118 (CNMF), 4.1423 (MTF-GLP), and 38.3981 (CNMF).

To visually demonstrate the pansharpened results, two reconstructed HS images with comparatively rich texture are selected from the testing data to make a visual comparison. Figs. 5 and 6 show the reconstructed results and absolute differences maps between the reconstructed results and the reference obtained by the compared methods. Obviously, the difference maps obtained by the proposed HPGAN achieve the smallest value difference. There is a big gap between the reconstructed results of the comparison method and the reference data at the boundary. The proposed HPGAN approach performs the best in reconstructing the boundary.

D. Component Analysis

This section analyzes the effects of major processing components, i.e., upsampling, HS high-frequency injection, PAN high-frequency injection, and deep details injection. Five quantitative indices used earlier are utilized to estimate $\mathbf{Y}_s$, $\mathbf{Y}_h$, $\mathbf{Y}_p$, and $\mathbf{Y}_g$. $\mathbf{Y}_s$ represents the upsampling HS image via interpolation. $\mathbf{Y}_h$, $\mathbf{Y}_p$, and $\mathbf{Y}_g$ denote $\mathbf{Y}_s$ injected with HS high-frequency details, $\mathbf{Y}_s$ injected with PAN high-frequency details, and the output of the generator, respectively. As listed in Table II, $\mathbf{Y}_g$ demonstrates the ability in spatial increasing with respect to CC, which means that spatial information of the HR PAN image is effectively injected into the LR HS image by HPGAN, similar to SAM, RMSE, ERGAS, and PSNR. Furthermore, $\mathbf{Y}_g$ has the best performance on all indices, which supports the conclusion that our network can increase the spatial information and keep the spectral information.

TABLE II
AVERAGE OBJECTIVE RESULTS ON THE CAVE DATA SET

Components	CC	SAM	RMSE	ERGAS	PSNR
$\mathbf{Y}_s$	0.9903	3.6744	0.0187	6.0275	32.0266
$\mathbf{Y}_p$	0.9722	7.2198	0.0247	10.3714	28.7684
$\mathbf{Y}_h$	0.9905	5.1457	0.0185	5.9082	32.0991
$\mathbf{Y}_g$	**0.9980**	**3.5866**	**0.0075**	**2.6265**	**39.4816**
Optimal Value	1	0	0	0	∞

E. Ablation Study

To objectively analyze the effects of each component in the proposed HPGAN, component analysis is carried out on 20 training HS images from the CAVE data sets. When analyzing the effects of an individual component, other components remain unchanged. In particular, Fig. 7 shows the average PSNR and SAM values obtained with/without BN under the same setting on network architecture. As shown in Fig. 7, PSNR that reflects the similarities between the pansharpened HS image and the reference image is a global index, and it shows an increasing trend when the epoch increases. Meanwhile, with BN, the PSNR values are consistently lower than those without BN. Note that the better pansharpening result means the smaller SAM. It can be observed from Fig. 7 that SAM with BN demonstrates a decreasing trend, and it exhibits consistently smaller than that without BN. In summary, one can see that the proposed HPGAN without BN is superior to that with BN.

The effectiveness of the adopted spectral constraint loss $l_{G^{(3)}}^{\mathrm{SAM}}$ is shown in Fig. 8, where SAM values with the spectral constraint loss are significantly better than that obtained by the 3-D generator network without the spectral constraint loss $l_{G^{(3)}}^{\mathrm{SAM}}$. The gradual improvement is relatively obvious, which means that $l_{G^{(3)}}^{\mathrm{SAM}}$ can enforce the final HR output close to the reference one. Therefore, the designed architecture for the proposed HPGAN model can boost pansharpening performance.

F. Generalization to New Data Sets

For the reconstructed results of the CAVE data set, both training and testing data come from the same sensor and cover the same spectral wavelength. It is a very challenging task to test the trained HPGAN model on the data set collected by different sensors, which can also validate the generalization of the model. Three additional data sets are used to test the trained HPGAN model on the CAVE data set. Among them, the Foster data set [36] has 33 spectral channels from 400–720 nm with 10 nm a band. The original size of each HS image in the Foster data set is $1341 \times 1022 \times 33$. In the experiment, all the HS images in the Foster data set are cropped to

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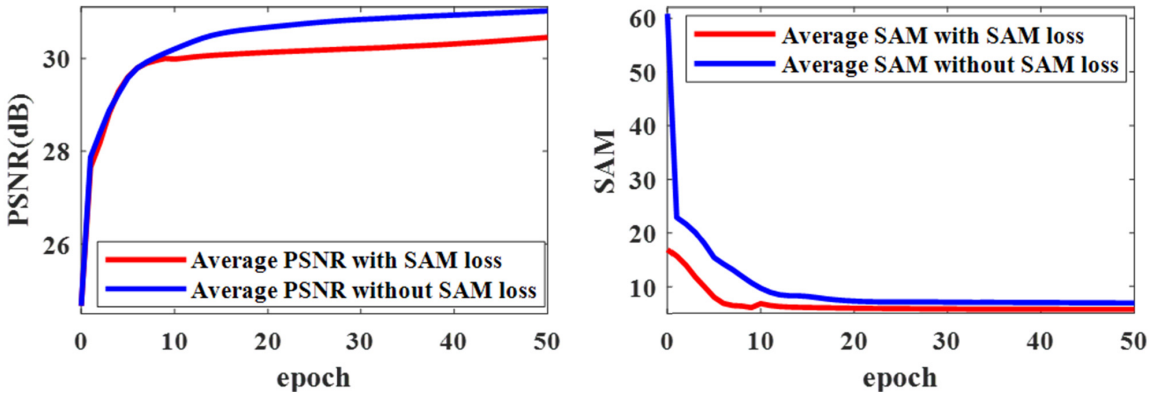


Fig. 8. Effects of the adopted spectral loss on PSNR and SAM with respect to epochs.

TABLE III
OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE FOSTER DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9868	2.4482	0.0050	4.3548	36.4075
MTF-GLP	0.9892	1.9138	0.0063	3.5070	35.5672
GFPCA	0.9858	2.2117	0.0059	4.6081	35.3982
HySure	0.9895	2.1663	0.0038	4.7814	38.6210
FUSE	0.9803	2.2449	0.0063	7.1166	35.3148
Lanaras	0.9779	3.4812	0.0058	6.0941	34.7697
CNMF	0.9890	2.3513	0.0041	4.0393	38.1665
DDLPS	0.9886	2.1117	0.0067	5.9939	34.8057
HPGAN	**0.9924**	**1.4736**	**0.0030**	**3.5726**	**41.0191**

TABLE IV
OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE PAVIA CENTER DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9086	7.3201	0.0444	5.7529	31.2387
MTF-GLP	0.9307	6.4335	0.0378	5.5357	30.0506
GFPCA	0.8768	8.0726	0.0517	7.3103	27.6115
HySure	0.9312	7.5287	0.0409	5.3905	31.2649
FUSE	0.9165	7.7916	0.0430	6.1227	29.3194
Lanaras	0.9249	7.1320	0.0400	5.2646	31.2392
CNMF	0.9362	6.0649	0.0366	5.1994	30.8376
DDLPS	0.9177	6.4275	0.0405	6.2443	28.9343
HPGAN	**0.9440**	**5.8349**	**0.0344**	**0.03447**	**32.1009**

512 × 512 in the spatial domain. The average evaluation indexes are reported in Table III. Since the CAVE and Foster data sets have a similar spectral range and consistent spectral resolution, the model trained on the CAVE data set is applied to the Foster data set, which also achieves satisfactory reconstruction results. Concretely, in terms of PSNR on average, HPGAN outperforms the GSA, MTF-GLP, GFPCA, HySure, FUSE, Lanaras, CNMF, and DDLPS by 4.61, 5.45, 5.62, 2.40, 5.70, 6.25, 2.85, and 6.21 dB, respectively. Besides, HPGAN achieves best on all the evaluation indexes, which illustrates that the trained HPGAN model on the CAVE data set can be better generalized to the Foster data set.

Fig. 9 presents the reconstructed HS images and the corresponding difference maps created by the compared methods. It is apparent that the difference values between the reconstructed and reference ones obtained by the MTF-GLP, GFPCA, FUSE, Lanaras, and DDLPS methods are relatively larger. The GSA, HySure, and CNMF methods can achieve better reconstruction results, but they are still not as good as our proposed HPGAN approach, especially in boundary and detailed reconstruction. These phenomena are fully consistent with the special, spatial, and global indexes in Table III.

Experiments have also been performed on the remote sensing HS images acquired by different sensors. A fractional of Pavia center data was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) over the city of Pavia, Italy. These data contain 115 spectral channels ranging from 430 to 860 nm, which is very different from the training data. For this data, 13 noisy channels are removed, and the remaining 102 bands are used for the experiment. The PAN image has 300 × 300 pixels with a spatial resolution of 1.3 m. The testing HS image has 75 × 75 pixels in the spatial domain with a spatial resolution of 5.2 m. As reported in Table IV, the proposed HPGAN model achieves the best results in terms of CC, SAM, RMSE, EGRAS, and PSNR. In particular, the average CC, SAM, RMSE, ERGAS, and PSNR obtained by the proposed HPGAN method all outperform the second-best approach.

Fig. 10 shows the false-color image of the pansharpened result for the fractional of Pavia center data. It can be seen that the results obtained by the GSA, MTF-GLP, GFPCA, Lanaras, CNMF, and DDLPS methods look blurry because the spatial information of the PAN image is not sufficiently injected. The boundaries in the pansharpened images obtained by the HySure and FUSE are too sharp in some areas. In summary, the color and details of the pansharpened HS image are close to that of the reference one, as indicated in Table IV.

The Chikusei data set is also used to evaluate the proposed model, which is captured by the Headwall's Hyperspec Visible and Near-Infrared, series C (VNIRC) imaging sensor over Chikusei, Ibaraki, Japan, on July 29, 2014. It contains 128 bands in the spectral range of 363–1018 nm. The PAN

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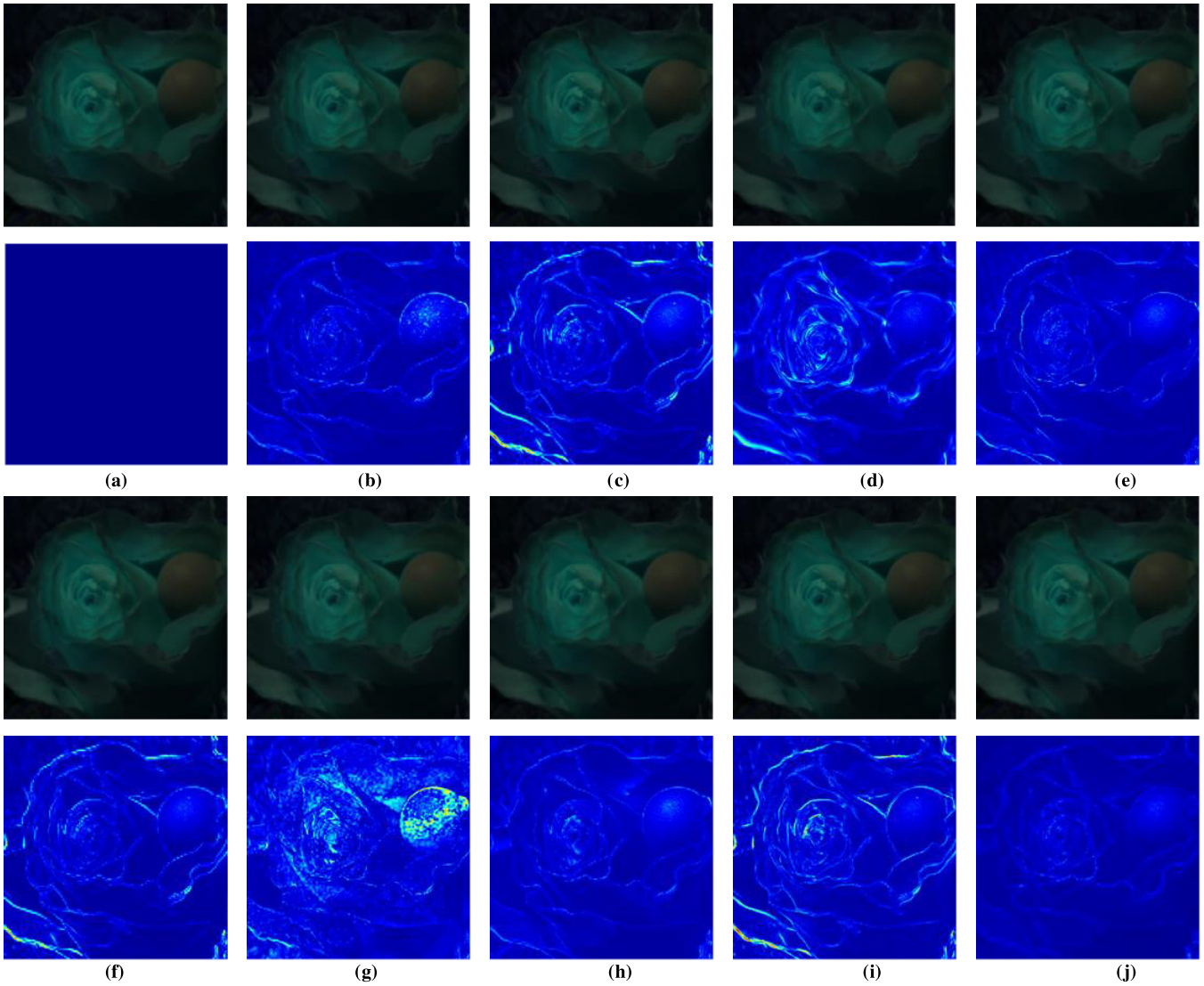


Fig. 9. Visual results obtained by different methods on an HS image with comparatively rich details from the Foster data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false-color image is chosen for clear visualization (red: 11, green: 22, and blue: 33). The even rows show the difference maps of the corresponding method.

image has 300×300 pixels with a spatial resolution of 2.5 m. The testing HS image has 75×75 pixels in the spatial domain with a spatial resolution of 10 m. Table V reports the performance of the compared methods on this remote sensing data set. The best results of CC, SAM, and RMSE indexes are provided by the proposed method with the default parameter settings, which proves that HPGAN provides better performance in spatial enhancement and spectral preservation. Besides, the performance will be improved when the parameters are specially optimized only for this data set.

The visual reconstructed results obtained by different methods for the Chikusei data set are shown in Fig. 11. For this data set, the GSA, MTF-GLP, FUSE, and Lanaras methods suffer from spectral distortion, the GFPCA and CNMF methods are over smooth, and the DDLPS method introduces redundant information.

Finally, the Moffett field data set is selected to validate the generalization of the trained model for the data that was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. This data set contains 224 bands in the spectral range of 400–2500 nm. After the water absorption

TABLE V

OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE CHIKUSEI DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9448	3.9458	0.0532	**4.2421**	**31.4454**
MTF-GLP	0.9378	4.1096	0.0528	5.3026	29.4774
GFPCA	0.8885	5.3771	0.0715	7.6942	26.3611
HySure	0.9361	4.7074	0.0608	4.8385	30.0765
FUSE	0.8860	6.1494	0.0820	6.4591	27.3643
Lanaras	0.9402	4.5123	0.0517	4.9337	30.0154
CNMF	0.9157	4.0742	0.0617	5.5231	29.0540
DDLPS	0.9316	4.151	0.0492	8.7823	28.7133
HPGAN	**0.9528**	**3.4967**	**0.0477**	4.6020	30.7059

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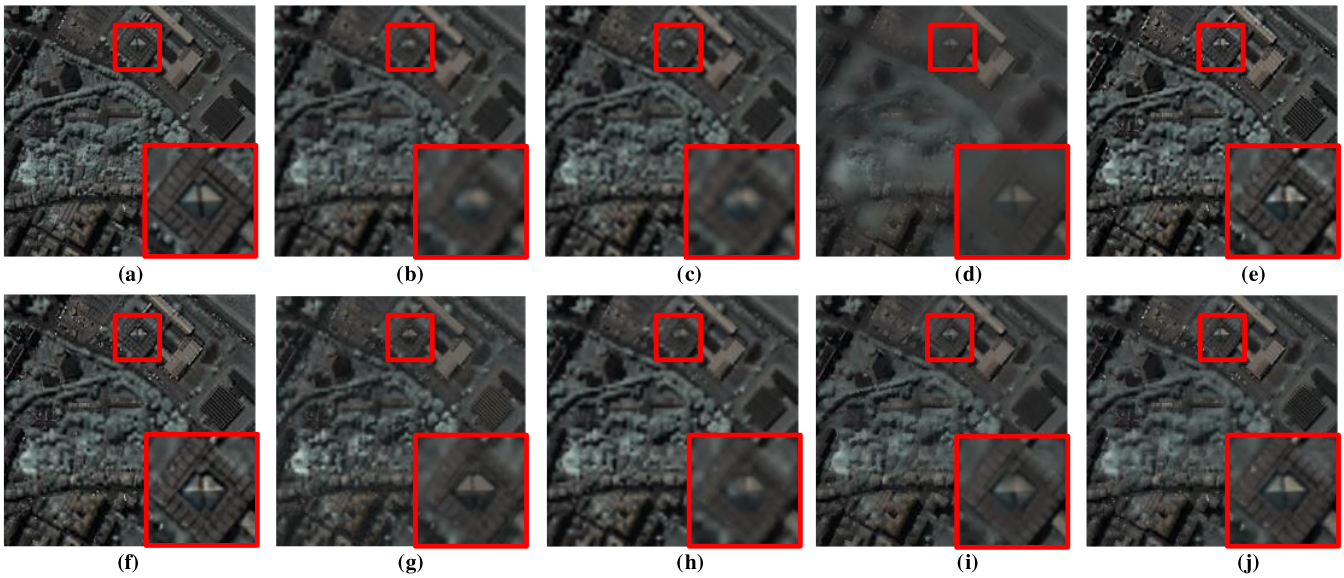


Fig. 10. Visual results obtained by different methods on the Pavia center data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false color image is chosen for clear visualization (red: 80, green: 90, and blue: 100).

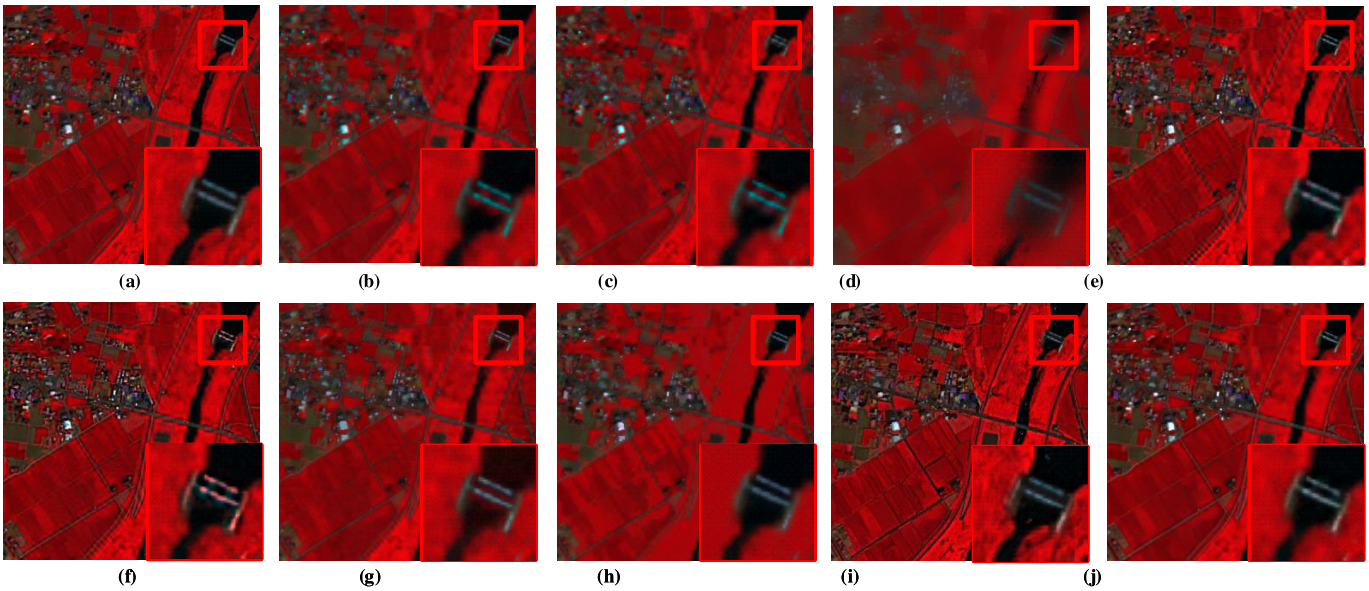


Fig. 11. Visual results obtained by different methods on the Chikusei data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false color image is chosen for clear visualization (red: 80, green: 60, and blue: 30).

and noise corrupted bands being removed, 176 bands are used for experimentation. The PAN image has 380×184 pixels with a spatial resolution of 50 m. The testing HS image has 95×46 pixels in the spatial domain with a spatial resolution of 200 m. The objective indexes are listed in Table VI. Obviously, the proposed HPGAN model yields the best values of CC, SAM, RMSE, ERGAS, and PSNR, which indicates that the proposed HPGAN model outperforms other methods in both spatial information injection and spectral preservation.

Fig. 12 shows the reconstruction results obtained by different methods for the Moffett field data set. Obviously, the structural information is better reconstructed by the proposed method. The GFPCA, Lanaras, CNMF, and DDLPS methods introduce redundant information resulting in spectral distortion because the color is different from the reference image. It is difficult to avoid halo artifacts and blurring problems induced by the GSA, MTF-GLP, HySure, and FUSE methods. We can conclude that the proposed HPGAN model is robust to different data sets captured by different sensors because it is trained on the high-frequency domain, and so the trained model on one data set can be better generalized to new data sets.

In addition, the running times of all the compared methods on all the testing HS images are listed in Table VII. All

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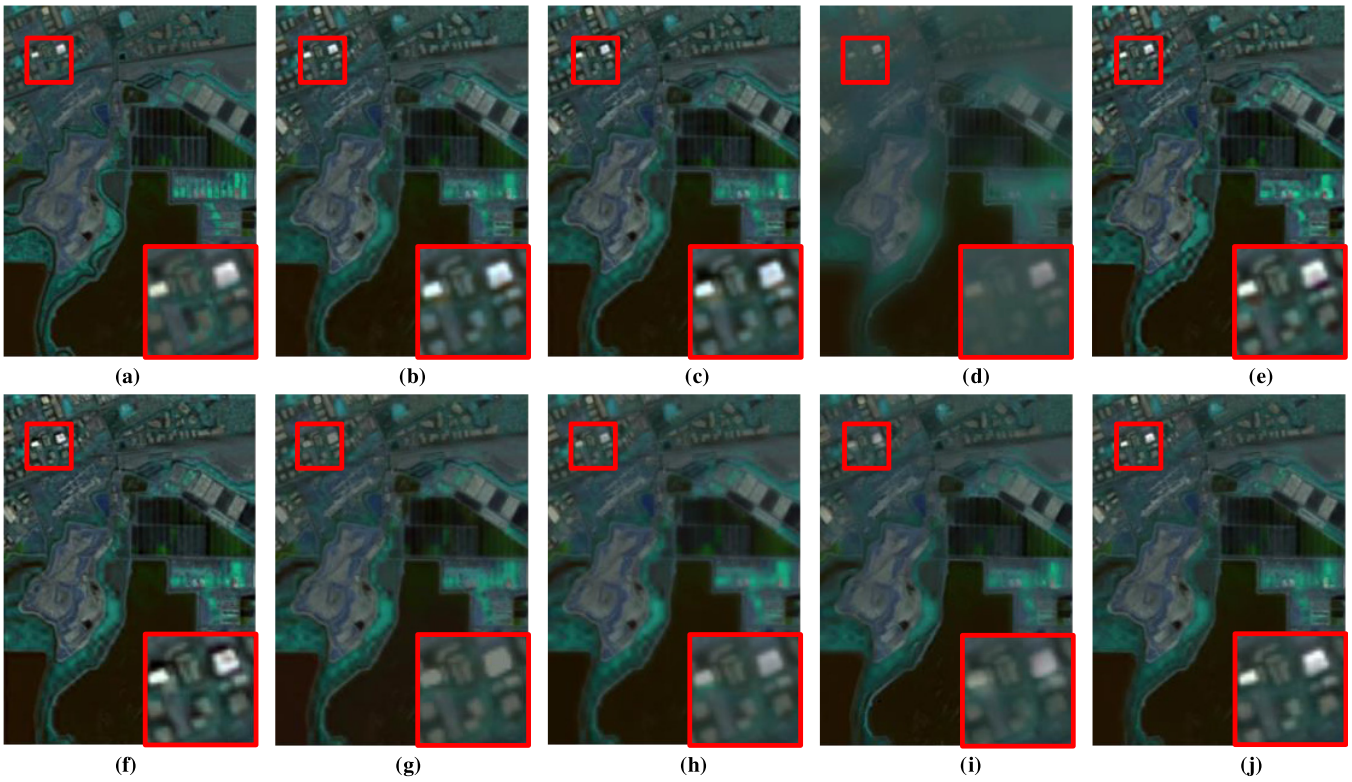


Fig. 12. Visual results obtained by different methods on the Moffett Field data set. (a) Ground truth. (b) GSA. (c) MTF-GLP. (d) GFPCA. (e) HySure. (f) FUSE. (g) Lanaras. (h) CNMF. (i) DDLPS. (j) HPGAN. Note that the false color image is chosen for clear visualization (red: 80, green: 90, and blue: 100).

TABLE VI

OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE MOFFETT FIELD DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9421	9.1223	0.0422	7.5676	29.5075
MTF-GLP	0.9556	7.1169	0.0352	6.4549	30.1287
GFPCA	0.9351	9.9160	0.0422	7.7145	28.3976
HySure	0.9549	7.5739	0.0369	6.6262	30.5805
FUSE	0.9551	7.8653	0.0356	6.4716	30.2367
Lanaras	0.9634	9.9452	0.0324	5.6799	31.2883
CNMF	0.9688	6.6941	0.0294	5.3664	31.9739
DDLPS	0.9584	6.4370	0.0326	6.1193	30.4353
HPGAN	**0.9729**	**5.7913**	**0.0277**	**5.0068**	**32.7793**

TABLE VII

RUNNING TIME (S) PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING FOUR DATA SETS

Method	CAVE	Foster	Pavia Center	Moffett field
GSA	**1.3298**	**1.4498**	1.5836	1.7712
MTF-GLP	9.7330	14.489	20.014	28.162
GFPCA	1.1681	1.0044	1.5116	2.0605
HySure	983.28	1216.7	356.56	281.27
FUSE	4.5152	4.1276	2.0280	2.7869
Lanaras	230.65	351.64	161.99	33.570
CNMF	16.966	20.509	13.853	15.692
DDLPS	2.8201	2.8581	**0.9495**	**0.7330**
HPGAN	12.819	12.861	19.626	25.562

the experiments are performed in MATLAB (R2014b) with Intel Core i3-3120 3.30-GHz CPU and TensorFlow 1.3.0 with NVIDIA Tesla K80 GPU. It can be observed that GSA performs the best for the CAVE and Foster data sets, while DDPLS is the most efficient one for the other two remote sensing images that contain hundreds of bands. Our proposed method is more time-consuming but significantly improves the objective evaluation indexes (PSNR, CC, RMSE, SAM, and ERGAS). It is worth noting that the proposed HPGAN method recovers the HR HS images with adaptively learned parameters, while the DDPLS method has to tune some parameters according to the final reconstruction performance.

G. Compare With 2-D Network

In this section, we compare the result between the 2-D network and our proposed 3-D network. We select the PanNet and perform experiments on the Moffett Field data set with fewer bands because 176 bands are too large for PanNet and the open-source code in PanNet chooses eight bands image to test. As shown in Fig. 13, the result of PanNet is too sharp. The result obtained by HPGAN is not as clear as PanNet but closer to the reference image. Furthermore, Table VIII reports the quality indices of the pansharpened results. From Table VIII, we can observe that the proposed method demonstrates better performance with 3-D convolutional layers than PanNet.

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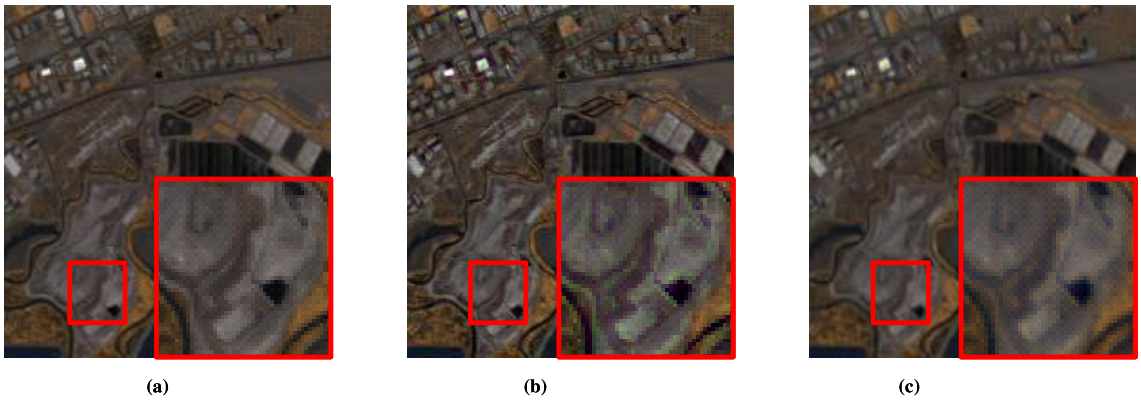


Fig. 13. Pansharpening images generated by PanNet and the proposed HPGAN. (a) Ground truth. (b) PanNet. (c) HPGAN.

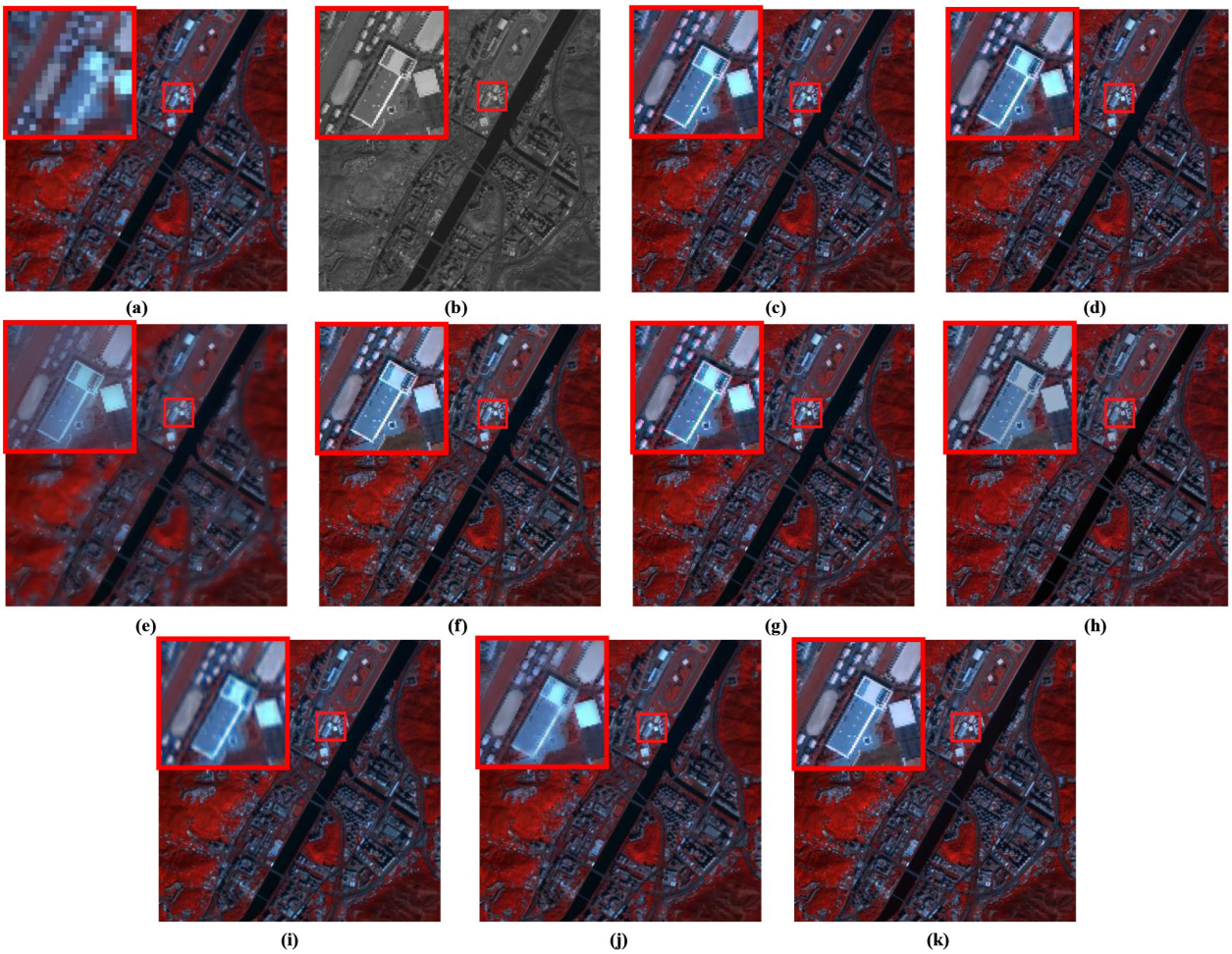


Fig. 14. Visual results obtained by different methods on the GF2 data set. (a) LR image. (b) HR PAN image. (c) GSA. (d) MTF-GLP. (e) GFPCA. (f) HySure. (g) FUSE. (h) Lanaras. (i) CNMF. (j) DDLPS. (k) HPGAN. Note that the false color image is chosen for clear visualization (red: 4, green: 2, and blue: 1).

H. Experiment on Real Data Set

The GF2 data set, which was acquired by the GaoFen-2 civilian optical remote sensing satellite, is used to evaluate the feasibility of our proposed HPGAN on a real data set. The GF-2 satellite was launched by China. Here, Gaofen means HR in Chinese. The real PAN image in this data set is of size 4000×4000, and the LR image is of size $1000 \times 1000 \times 4$ in the spectral range of 450–890 nm. The original images as well as the pansharpened results are displayed in Fig. 14. As shown in Fig. 14, the pansharpened result obtained by HPGAN can properly enhance spatial resolution while finely preserving the structure underlying the LR image. Since the HR image of the real data set is not available, we use the same scale to degrade the original images for objective evaluation according to Wald's protocol [7], [8]. Table IX lists the objective indexes

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TABLE VIII

QUANTITATIVE RESULTS OF VARIOUS NETWORKS IN THE EXPERIMENT USING THE MOFFETT FIELD DATA SET WITH FEWER BANDS

Network	CC	SAM	RMSE	ERGAS	PSNR
PanNet	0.9616	4.6577	0.0281	4.1205	31.1706
HPGAN	**0.9793**	**2.6039**	**0.0170**	**2.5197**	**35.6540**

TABLE IX

OBJECTIVE EVALUATION INDEXES PRODUCED BY VARIOUS METHODS IN THE EXPERIMENT USING THE GF2 DATA SET

Method	CC	SAM	RMSE	ERGAS	PSNR
GSA	0.9209	6.6017	0.0542	7.8876	25.7128
MTF-GLP	0.9165	6.4970	0.0562	8.1618	25.3874
GFPCA	0.9008	6.9574	0.0568	9.0872	25.0170
HySure	0.8983	7.9278	0.0614	9.7792	24.3408
FUSE	0.9156	7.3318	0.0584	8.2842	25.2298
Lanaras	0.8977	8.1697	0.0613	9.8785	24.3263
CNMF	0.9122	6.2259	0.0583	8.9223	24.8442
DDLPS	0.9370	6.1716	0.0452	7.1406	27.0174
HPGAN	**0.9525**	**3.9086**	**0.0401**	**6.2055**	**28.0952**

of the comparison methods. It can be seen that HPGAN ranks first for five evaluation indexes, which further illustrates the superior performance of the proposed method in spatial SR and spectral preservation.

V. CONCLUSION

In this article, we propose a novel 3DGAN for the HS pansharpening (HPGAN) method. The 3-D high-frequency block is generated to train two images with different dimensions in a unique framework, which also reduces the sensitivity to different sensor data. The induced 3-D spectral–spatial generator network considers global constraint, spectral constraint, and spatial constraint to superresolve HS images while preserving spectral information. Based on the analysis of the characteristics of HS imaging, a PAN discriminator network is developed to distinguish generated PAN images from the real one rather than to distinguish the generated HR HS image from the reference HR HS image. Extensive experimental results on several data sets indicate that the proposed HPGAN not only produces more favorable reconstruction but also generalizes better to new data sets compared with other state-of-the-arts in terms of subjective and objective evaluation. In the future, how to generalize the proposed method in a practical situation and consider some ways to avoid overfitting will be the focus.

In the future, we will focus on designing loss function according to the characteristics of different spectral ranges to reduce spectral distortion, paying particular attention to spectral ranges that are not the same as those of the PAN image.

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Weiying Xie (Member, IEEE) received the B.S. degree in electronic information science and technology from the University of Jinan, Jinan, China, in 2011. She received the M.S. and Ph.D. degrees in communication and information systems, Lanzhou University, Lanzhou, China, in 2014 and Xidian University, Xi'an, China, in 2017, respectively.

She is an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 30 articles in refereed journals, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the *Neural Networks*, and the *Pattern Recognition*. Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.



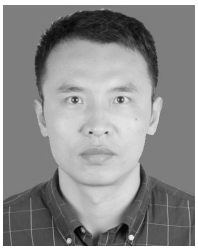
Yuhang Cui received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2018. He is pursuing the M.S. degree with the Image Coding and Processing Center at State Key Laboratory of Integrated Services Networks, Xidian University.

His research interests include image processing, hyperspectral image superresolution, and deep learning.



Yunsong Li received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joins the School of Telecommunications Engineering, Xidian University, in 1999, where he is a Professor. Prof. Li is the director of the image coding and processing center at the State Key Laboratory of Integrated Services Networks. His research interests focus on image and video processing and high-performance computing.



Jie Lei (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2006 and 2010, respectively.

He has been a Visiting Scholar at the Department of Computer Science of University of California, Los Angeles, CA, USA, from 2014 to 2015. He is an Associate Professor at the School of Telecommunications Engineering, Xidian University, and is a member of the Image Coding and Processing Center at the State Key Laboratory of Integrated Services Networks. His research interests focus on image and video processing, computer vision, and customized computing for big-data applications.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. She is also an Adjunct Professor with the College of Surveying and Geo-informatics, Tongji University, Shanghai, China. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a Fellow of the IEEE-Institute of Electrical and Electronics Engineers, and a Fellow of the SPIE-International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was a Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013, and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She has served as an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She was the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, China, in 2012.



Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively. She was an exchange Ph.D. Student of Mississippi State University, Starkville, MS, USA.

She is a Post-Doctoral Researcher and a Lecturer with the School of Telecommunication, Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.

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